Screen these results against the NEPM

Location Code,Depth,Sampled Date,Chem Name,Prefix,Result,Unit
BH01,3 - 3.1,15/11/2023,Arsenic,,45,mg/kg
BH01,3 - 3.1,15/11/2023,Cadmium,,24,mg/kg
BH01,3 - 3.1,15/11/2023,Chromium,,615,mg/kg
BH01,3 - 3.1,15/11/2023,Copper,,253,mg/kg
BH01,3 - 3.1,15/11/2023,Lead,,1800,mg/kg
BH01,3 - 3.1,15/11/2023,Mercury,,0.1,mg/kg
BH01,3 - 3.1,15/11/2023,Nickel,,694,mg/kg
BH01,3 - 3.1,15/11/2023,Zinc,,2340,mg/kg
BH01,4.8 - 4.9,15/11/2023,Arsenic,,26,mg/kg
BH01,4.8 - 4.9,15/11/2023,Cadmium,,5,mg/kg
BH01,4.8 - 4.9,15/11/2023,Chromium,,50,mg/kg
BH01,4.8 - 4.9,15/11/2023,Copper,,2150,mg/kg
BH01,4.8 - 4.9,15/11/2023,Lead,,837,mg/kg
BH01,4.8 - 4.9,15/11/2023,Mercury,,0.2,mg/kg
BH01,4.8 - 4.9,15/11/2023,Nickel,,88,mg/kg
BH01,4.8 - 4.9,15/11/2023,Zinc,,5310,mg/kg
BH03,0.5 - 0.6,15/11/2023,Arsenic,,11,mg/kg
BH03,0.5 - 0.6,15/11/2023,Cadmium,<,1,mg/kg
BH03,0.5 - 0.6,15/11/2023,Chromium,,42,mg/kg
BH03,0.5 - 0.6,15/11/2023,Copper,,23,mg/kg
BH03,0.5 - 0.6,15/11/2023,Lead,,22,mg/kg
BH03,0.5 - 0.6,15/11/2023,Mercury,<,0.1,mg/kg
BH03,0.5 - 0.6,15/11/2023,Nickel,,37,mg/kg
BH03,0.5 - 0.6,15/11/2023,Zinc,,72,mg/kg
BH03,2 - 2.1,15/11/2023,Arsenic,,6,mg/kg
BH03,2 - 2.1,15/11/2023,Cadmium,<,1,mg/kg
BH03,2 - 2.1,15/11/2023,Chromium,,28,mg/kg
BH03,2 - 2.1,15/11/2023,Copper,,49,mg/kg
BH03,2 - 2.1,15/11/2023,Lead,,184,mg/kg
BH03,2 - 2.1,15/11/2023,Mercury,<,0.1,mg/kg
BH03,2 - 2.1,15/11/2023,Nickel,,26,mg/kg
BH03,2 - 2.1,15/11/2023,Zinc,,225,mg/kg
BH05,1.8 - 1.9,15/11/2023,Arsenic,,5,mg/kg
BH05,1.8 - 1.9,15/11/2023,Cadmium,<,1,mg/kg
BH05,1.8 - 1.9,15/11/2023,Chromium,,56,mg/kg
BH05,1.8 - 1.9,15/11/2023,Copper,,22,mg/kg
BH05,1.8 - 1.9,15/11/2023,Lead,,26,mg/kg
BH05,1.8 - 1.9,15/11/2023,Mercury,<,0.1,mg/kg
BH05,1.8 - 1.9,15/11/2023,Nickel,,55,mg/kg
BH05,1.8 - 1.9,15/11/2023,Zinc,,344,mg/kg
BH05,5 - 5.1,15/11/2023,Arsenic,,12,mg/kg
BH05,5 - 5.1,15/11/2023,Cadmium,,2,mg/kg
BH05,5 - 5.1,15/11/2023,Chromium,,34,mg/kg
BH05,5 - 5.1,15/11/2023,Copper,,1460,mg/kg
BH05,5 - 5.1,15/11/2023,Lead,,557,mg/kg
BH05,5 - 5.1,15/11/2023,Mercury,<,0.1,mg/kg
BH05,5 - 5.1,15/11/2023,Nickel,,80,mg/kg
BH05,5 - 5.1,15/11/2023,Zinc,,30000,mg/kg
BH05,7.5 - 7.6,15/11/2023,Arsenic,<,5,mg/kg
BH05,7.5 - 7.6,15/11/2023,Cadmium,<,1,mg/kg
BH05,7.5 - 7.6,15/11/2023,Chromium,,52,mg/kg
BH05,7.5 - 7.6,15/11/2023,Copper,,43,mg/kg
BH05,7.5 - 7.6,15/11/2023,Lead,,68,mg/kg
BH05,7.5 - 7.6,15/11/2023,Mercury,<,0.1,mg/kg
BH05,7.5 - 7.6,15/11/2023,Nickel,,79,mg/kg
BH05,7.5 - 7.6,15/11/2023,Zinc,,1950,mg/kg

To screen these results against the **NEPM (ASC NEPM 2013, amended 2019)**, I'll assume a **generic residential scenario (HIL A)** unless you tell me otherwise. If this is a different land use (e.g. commercial/industrial HIL D), let me know and I'll re-run it.

### Adopted NEPM HIL A criteria (mg/kg)

- Arsenic: 20
- Cadmium: 20
- Chromium (total): 1000 *(note: Cr VI has a much lower limit but not specified here)*
- Copper: 3000
- Lead: 300
- Mercury (inorganic): 1
- Nickel: 600
- Zinc: 7400

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## Screening Results Summary

### Exceedances (above HIL A)

#### BH01 – 3–3.1 m

- Arsenic: **45** ↑
- Cadmium: **24** ↑
- Lead: **1800** ↑
- Nickel: **694** ↑

👉 **Multiple significant exceedances**

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#### BH01 – 4.8–4.9 m

- Arsenic: **26** ↑
- Lead: **837** ↑

👉 **Moderate contamination (primarily Pb)**

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#### BH05 – 5–5.1 m

- Zinc: **30,000** ↑ (very high)

👉 **Major zinc hotspot**

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#### BH05 – 7.5–7.6 m

- Nickel: **79** ❌ (below 600 → OK)
- Zinc: **1950** ❌ (below 7400 → OK)

👉 **No exceedances here**

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### Compliant (below HIL A)

All of the following intervals are **below HIL A**:

- BH03 (both depths)
- BH05 (1.8–1.9 m)
- BH05 (7.5–7.6 m)
- Most analytes at BH01 except those noted above

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## Key Observations

### 1. Lead contamination is the main issue

- BH01 shows **very high Pb (up to 1800 mg/kg)**

- Exceeds HIL A by **~6×**

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### 2. Nickel hotspot

- BH01 (3–3.1 m): **694 mg/kg** (>600)
- Could be natural or anthropogenic—needs context

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### 3. Extreme zinc at BH05 (5 m)

- **30,000 mg/kg (~4× HIL A)**
- Likely **point source contamination**

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### 4. Cadmium slightly elevated

- Only one exceedance (BH01 shallow deep zone)
- Not widespread

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### 5. Chromium, copper, mercury

- All **well below criteria**
- Chromium is high (615 mg/kg) but still <1000

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## Important Technical Notes

- **Depth matters:**
  Many exceedances occur **>3 m depth**, which may reduce risk depending on exposure scenario.
- **Lead is risk-driving:**
  Even one shallow exceedance can trigger remediation in residential settings.
- **Zinc spike (30,000 mg/kg):**
  This is unusually high and may indicate:
  - Fill material
  - Industrial waste
  - Localised contamination source
- **"<" values:**
  Treated as **below detection** and compliant.

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## Conclusion

- **Contamination is present but localized**
- **Primary contaminants of concern:**
  - Lead (BH01)
  - Nickel (BH01)
  - Zinc (BH05 deeper zone)
- **BH03 and most of BH05 are clean**

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## Next Steps (Typical)

Depending on project stage:

- Refine **conceptual site model (CSM)**
- Check **depth to exposure** (are these soils accessible?)
- Consider:
  - HIL D screening (if industrial)
  - Ecological criteria (EILs)
  - Groundwater impacts

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If you want, I can:

- Re-screen using **HIL B/C/D**
- Add **EIL comparison**
- Flag **risk by exposure pathway**
- Or produce a **report-ready table** 👍